EIGENVALUE CONVERGENCE FOR POLYNOMIALS IN INDEPENDENT GUE MATRICES

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Abstract. We prove that the empirical eigenvalue distribution of any fixed scalar polynomial in independent normalized GUE matrices converges weakly in probability to the Brown measure of the same polynomial in free semicircular elements. The proof uses a polynomial lower bound, with probability tending to one, for the least singular value of a fixed monic matrix pencil in the GUE inputs.

1. Introduction

Let $d \geq 1$ and let $H^{(N)} := (H_1^{(N)}, \dots, H_d^{(N)})$ consist of independent $N \times N$ matrices from the Gaussian unitary ensemble (GUE), each with density proportional to $\exp(-N \operatorname{Tr}(H^2)/2)$ on the real vector space of Hermitian matrices. Here Tr denotes the unnormalized matrix trace and $\operatorname{tr}_N := N^{-1} \operatorname{Tr}$. Let $s := (s_1, \dots, s_d)$ be a free centered semicircular family of variance one in a finite von Neumann algebra $\mathcal{M}$ with faithful normal tracial state τ. Each s_j has spectral density $(2\pi)^{-1}\sqrt{4 - x^2}$ on $[-2, 2]$. Freeness means that the trace of a product of centered elements from the unital algebras generated by the individual s_j vanishes whenever consecutive factors come from distinct algebras [NS06, Definition 2.16, Remark 2.17(2), and Definition 5.3].

For each scalar polynomial Q in the input variables, $\operatorname{tr}_N(Q(H^{(N)}))$ converges in probability to $\tau(Q(s))$, and $\max_j \|H_j^{(N)}\|$ converges in probability to 2 [CGP22, pp. 1–2 of arXiv v1]. For a self-adjoint scalar polynomial P, this gives convergence of the empirical eigenvalue distribution of $P(H^{(N)})$ to the spectral distribution of $P(s)$. For a general polynomial, the eigenvalues can lie in the complex plane, and the corresponding limit is given by the Brown measure.

We use the Brown measure convention of [MS17, Chapter 11, Theorem 5, Definition 6 and pp. 268–270].

Definition 1.1. Let $\mathcal{M}$ be a finite von Neumann algebra with faithful normal tracial state τ, and let $a \in \mathcal{M}$. The Brown measure of a is the measure μ_a^{Br} given by

$$\mu_a^{\mathrm{Br}} := \frac{1}{2\pi} \Delta u_a, \qquad u_a(z) := \tau(\log |a - z1_{\mathcal{M}}|).$$

Here $|b| := (b^*b)^{1/2}$, the trace of the logarithm is the spectral integral with respect to τ, and Δ acts in the sense of distributions on $\mathbb{C}$.

The potential u_a is locally integrable, and μ_a^{Br} is a probability measure supported in the spectrum of a. Moreover, $u_a(z) = \int_{\mathbb{C}} \log |z - w| \, d\mu_a^{\mathrm{Br}}(w)$, and for normal a the Brown measure agrees with its spectral distribution.

Belinschi, Śniady and Speicher [BŚS18, Section 7] formulate the eigenvalue convergence conjecture for general polynomials in independent self-adjoint Gaussian matrices. Han [Han26, Theorem 1.2] proves weak convergence in probability for polynomials in independent complex Ginibre matrices using only the unstarred variables, with the corresponding free circular limit. Two independent GUE inputs can be combined as $(H_1^{(N)} + \mathrm{i}H_2^{(N)})/\sqrt{2}$ to produce a normalized complex Ginibre matrix, whose real and imaginary entry coordinates are independent centered Gaussians of variance $1/(2N)$. Writing a general GUE polynomial in these variables introduces

their adjoints, a case discussed in [Han26, Question 1.8]. We prove weak convergence in probability for the full class of GUE polynomials.

Theorem 1.2. *Let $d \geq 1$ and fix $P \in \mathbb{C}\langle x_1, \ldots, x_d \rangle$, where $x_1, \ldots, x_d$ are self-adjoint indeterminates. Let $H^{(N)}$ be as above, and let $s_1, \ldots, s_d \in \mathcal{M}$ be freely independent centered semicircular elements of variance one in a finite von Neumann algebra with faithful normal tracial state τ. Then the empirical measure*

$$\mu_{P(H^{(N)})} := \frac{1}{N} \sum_{k=1}^{N} \delta_{\lambda_k(P(H^{(N)}))}$$

converges weakly in probability to $\mu^{\mathrm{Br}}_{P(s)}$. The eigenvalues are counted with algebraic multiplicity, and the convergence holds against every bounded continuous real-valued test function on $\mathbb{C}$.

The main estimate concerns a matrix pencil with identity constant term. For a Hermitian tuple $H = (H_1, \ldots, H_d)$ of size N, let

$$T_N(H) := I_m \otimes I_N + \sum_{j=1}^{d} R_j \otimes H_j, \tag{1.1}$$

where the coefficient matrices $R_1, \ldots, R_d \in M_m(\mathbb{C})$ are fixed. We call such a pencil monic. For a square matrix B, write $s_{\min}(B) := \min_{\|v\|=1} \|Bv\|$.

Theorem 1.3. *Let $d, m \geq 1$, and let $R_1, \ldots, R_d \in M_m(\mathbb{C})$ be arbitrary fixed matrices. There exists $A > 0$ such that*

$$\mathbb{P}\{s_{\min}(T_N(H^{(N)})) < N^{-A}\} \longrightarrow 0. \tag{1.2}$$

For each $M > 0$ there exist $C, \alpha, \varepsilon_0, c > 0$, independent of N, such that for all sufficiently large N and $0 < \varepsilon \leq \varepsilon_0$,

$$\begin{aligned} &\mathbb{P}\{s_{\min}(T_N(H^{(N)})) \leq \varepsilon,\ \max_j \|H_j^{(N)}\| \leq M\} \\ &\qquad \leq CN^{5/2}\varepsilon^{\alpha} \exp(CN^{5/2}\varepsilon^{\alpha}) + e^{-cN^2}. \end{aligned} \tag{1.3}$$

The first step is to show that a small singular value puts the Hermitian input tuple close to a tuple for which the pencil is singular. In Section 2, we compress an approximate kernel vector and the action of the inputs on its components to a space of dimension bounded in terms of m and d. A polynomial distance inequality in that dimension yields a nearby exact kernel equation, which we lift to the original space. Section 3 then bounds the Gaussian measure of a neighborhood of the zero set of the determinant. The determinant has degree at most mN, which gives a polynomial dependence on N in the exponent and prefactor of the tube bound (1.3).

In Section 4, a Schur complement transfers Theorem 1.3 to $P(H^{(N)}) - zI_N$ for fixed $z \neq P(0)$. We use hermitization to pass from the squared singular values to eigenvalue measures, following the logarithmic truncation argument of [Han26, Section 5]. The pencil estimate controls the logarithm at the bottom of the singular-value distribution, and a comparison of Cauchy transforms controls the number of singular values at larger scales.

2. Distance to singular pencils

Proposition 2.1. *Fix $R_1, \ldots, R_d \in M_m(\mathbb{C})$ and $M > 0$. For a tuple $H = (H_1, \ldots, H_d)$ of Hermitian $N \times N$ matrices, use the pencil $T_N(H)$ from (1.1). There exist $C, \alpha, \varepsilon_0 > 0$ such that, for every N and every such H with $\max_j \|H_j\| \leq M$, the following holds. If $s_{\min}(T_N(H)) \leq \varepsilon \leq \varepsilon_0$*

with $\varepsilon > 0$, there are Hermitian $N \times N$ matrices $\Delta H_1, \dots, \Delta H_d$ such that $T_N(H + \Delta H)$ is singular and

$$\left(\sum_{j=1}^{d} \|\Delta H_j\|_{\mathrm{HS}}^2 \right)^{1/2} \leq C\varepsilon^{\alpha}. \tag{2.1}$$

Here $\|X\|_{\mathrm{HS}} := (\mathrm{Tr}(X^*X))^{1/2}$ is the Hilbert–Schmidt norm, with the unnormalized trace on the appropriate matrix size.

Proof. Choose a unit vector $v = (v_1, \dots, v_m) \in \mathbb{C}^m \otimes \mathbb{C}^N$ with $\|T_N(H)v\| \leq \varepsilon$. Let $W := \mathrm{span}\{v_1, \dots, v_m\}$, and let $W_1 := \sum_{j=1}^{d} P_{W^\perp} H_j W$, where $P_{W^\perp}$ is the orthogonal projection onto $W^\perp$. Writing $k := \dim W$ and $\ell := \dim W_1$, we have $1 \leq k \leq m$ and $0 \leq \ell \leq dk$. Choose isometries $U : \mathbb{C}^k \to \mathbb{C}^N$ and $V : \mathbb{C}^\ell \to \mathbb{C}^N$ onto W and W_1, respectively, with $\mathbb{C}^0 = \{0\}$. For each j put $A_j := U^* H_j U$ and $B_j := V^* H_j U$, so that A_j is Hermitian and $B_j \in M_{\ell,k}(\mathbb{C})$. By the definition of W_1,

$$H_j U = U A_j + V B_j. \tag{2.2}$$

Write $w := (U^*v_1, \dots, U^*v_m) \in \mathbb{C}^m \otimes \mathbb{C}^k$. The two coordinate vectors of $T_N(H)v$ in the orthogonal summands $\mathbb{C}^m \otimes W$ and $\mathbb{C}^m \otimes W_1$ are

$$w + \sum_{j=1}^{d} (R_j \otimes A_j) w \quad \text{and} \quad \sum_{j=1}^{d} (R_j \otimes B_j) w. \tag{2.3}$$

Equation (2.2) gives no component outside these summands, so the sum of the squared norms in (2.3) is $\|T_N(H)v\|^2$.

Use Hilbert–Schmidt orthonormal real coordinates for each A_j, and split the entries of each B_j and of w into real and imaginary parts. For each (k, ℓ) in the indicated ranges, these choices give $p_{k,\ell} := dk^2 + 2d\ell k + 2mk$ real coordinates for $x := (A, B, w)$. Let $F_{k,\ell} : \mathbb{R}^{p_{k,\ell}} \to \mathbb{R}^{2m(k+\ell)+1}$ be the polynomial map whose coordinates are the real and imaginary parts of (2.3) and the scalar $\|w\|^2 - 1$. The map is quadratic and depends only on the coefficients R_j and the pair (k, ℓ). Since $\|w\| = 1$ and $\|A_j\|_{\mathrm{HS}}, \|B_j\|_{\mathrm{HS}} \leq \sqrt{k}M$, the Euclidean norms of x and its image satisfy

$$|x|^2 \leq 2dkM^2 + 1, \qquad |F_{k,\ell}(x)| \leq \varepsilon. \tag{2.4}$$

The real polynomial distance inequality of Kurdyka and Spodzieja, as stated in [KSS16, p. 2, (KS2)], gives the following bound for a polynomial map $F : \mathbb{R}^p \to \mathbb{R}^q$ of degree $D \geq 1$ with a nonempty zero set. For some $c_F > 0$ and every $y \in \mathbb{R}^p$,

$$|F(y)| \geq c_F \left(\frac{\mathrm{dist}(y, F^{-1}(0))}{1 + |y|^2} \right)^{D(6D-3)^{p-1}}.$$

When $F_{k,\ell}$ has a zero, its zero set is closed, so the distance from x is attained at some $x' := (A', B', w')$. Using $D = 2$ and (2.4) gives

$$|x' - x| \leq C_{k,\ell} \varepsilon^{\alpha_{k,\ell}}, \qquad \alpha_{k,\ell} := (2 \cdot 9^{p_{k,\ell}-1})^{-1}, \tag{2.5}$$

where $C_{k,\ell}$ depends only on the map and M. If the zero set is empty, then $|F_{k,\ell}|$ has a positive minimum on the closed ball $|x|^2 \leq 2dkM^2 + 1$. There are only finitely many pairs (k, ℓ), so we can choose $\varepsilon_0 \leq 1$ below these positive minima, and choose a common exponent $\alpha > 0$ and a constant C_1 in (2.5).

With the original U and V, define

$$\Delta H_j := U(A_j' - A_j)U^* + V(B_j' - B_j)U^* + U(B_j' - B_j)^* V^*.$$

This matrix is Hermitian and satisfies $(H_j + \Delta H_j)U = UA_j' + VB_j'$ by (2.2) and $V^*U = 0$. The coordinates of $T_N(H + \Delta H)(I_m \otimes U)w'$ are therefore the expressions in (2.3) with A_j, B_j, w replaced

by A'_j, B'_j, w'. Both expressions vanish because $F_{k,\ell}(x') = 0$, and its normalization coordinate gives $\|(I_m \otimes U)w'\| = 1$. This produces a kernel vector even if its components span a proper subspace of W. The three blocks in ΔH_j are pairwise Hilbert–Schmidt orthogonal, so

$$\sum_{j=1}^{d} \|\Delta H_j\|_{\mathrm{HS}}^2 = \sum_{j=1}^{d} \left(\|A'_j - A_j\|_{\mathrm{HS}}^2 + 2\|B'_j - B_j\|_{\mathrm{HS}}^2\right) \le 2|x' - x|^2.$$

This proves (2.1) with $C = \sqrt{2}C_1$. □

3. Gaussian tubes and the least singular value

Proposition 2.1 reduces the small singular value event to proximity to a real polynomial zero set. We first bound the Gaussian measure of the points within a given distance of such a set.

Lemma 3.1. *Let q be a nonzero, nonnegative real polynomial on $\mathbb{R}^n$ of degree at most D, where $D \ge 1$, and suppose $Z := q^{-1}(0)$ is nonempty. If g is a standard Gaussian vector in $\mathbb{R}^n$, then for every $\rho > 0$,*

$$\mathbb{P}\{\operatorname{dist}(g, Z) \le \rho\} \le CD\sqrt{n}\,\rho \exp\bigl(C(D+1)\sqrt{n}\,\rho\bigr) + e^{-cn}$$

for universal constants $C, c > 0$.

Proof. Let X be uniform in the ball of radius $\sigma > 0$ centered at the origin in $\mathbb{R}^n$. For a smooth hypersurface Y given by a regular level of a polynomial of degree at most D, the codimension-one case of Lotz's tube estimate [Lot15, Theorem 1.1] gives

$$\mathbb{P}\{\operatorname{dist}(X, Y) \le \rho\} \le 4\sum_{r=1}^{n} \binom{n}{r} (2D\rho/\sigma)^r (1 + \rho/\sigma)^{n-r}.$$

Set $u := \rho/\sigma$, so the right side is $4((1 + (2D+1)u)^n - (1+u)^n)$. By the mean value theorem, this expression is bounded by

$$\frac{8nD\rho}{\sigma} \exp\left(\frac{n(2D+1)\rho}{\sigma}\right). \tag{3.1}$$

To apply this bound to Z, fix $\eta > 0$ and consider the zeros of q in the closed ball of radius $\sigma + \rho$. Since Z is closed, every point of the tube in the ball of radius σ has a nearest zero in this larger ball. If there are no such zeros, the tube event in the ball of radius σ is empty. Otherwise, near each such zero y there is a point p_y with $|p_y - y| < \eta/2$ and $q(p_y) > 0$, since a nonzero nonnegative polynomial cannot vanish on an open ball. By compactness, finitely many open balls of radius $\eta/2$ centered at these zeros cover the relevant zero set. Choose a positive regular value δ of q less than all the values $q(p_y)$ for the chosen centers, using Sard's theorem. For each relevant zero y', the segment from y' to the positive point associated with a covering ball lies within distance η of y' and meets $q^{-1}(\delta)$. Every point in the ball of radius σ within distance ρ of Z is therefore within distance $\rho + \eta$ of $q^{-1}(\delta)$. The latter is a smooth hypersurface defined by a polynomial of degree at most D. Applying (3.1) with $\rho + \eta$ and letting η decrease to zero gives (3.1) for Z.

For the Gaussian estimate, first sample $S > 0$ with the chi distribution with $n+2$ degrees of freedom and then sample a point uniformly in the ball of radius S. With ω_n the volume of the unit ball in $\mathbb{R}^n$, the resulting density at $x \in \mathbb{R}^n$ is

$$\int_{|x|}^{\infty} \frac{1}{\omega_n s^n} \frac{s^{n+1} e^{-s^2/2}}{2^{n/2}\Gamma(n/2+1)}\, ds = (2\pi)^{-n/2} e^{-|x|^2/2}.$$

Since S^2 is the sum of $n+2$ independent squared standard normal variables, the exponential Markov bound gives $\mathbb{P}\{S < \sqrt{n}/2\} \le e^{n/8} 2^{-(n+2)/2} \le e^{-cn}$ for a universal $c > 0$. On $S \ge \sqrt{n}/2$, the bound (3.1) for Z has prefactor at most $16D\sqrt{n}\,\rho$ and exponent at most $2(2D+1)\sqrt{n}\,\rho$. Together these estimates give the claim. □

Proof of Theorem 1.3. Write each component of $H^{(N)}$ in a real Hilbert–Schmidt orthonormal basis of Hermitian matrices. Its law in these coordinates is that of $N^{-1/2}g$, where g is standard Gaussian in $\mathbb{R}^n$ and $n := dN^2$. Indeed, $\operatorname{Tr}(H_j^2)$ is the squared Euclidean norm of the coordinate vector. Using the same coordinates to evaluate the pencil, define

$$q_N(x) := \left|\det T_N(N^{-1/2}x)\right|^2, \qquad Z_N := q_N^{-1}(0).$$

The determinant has degree at most mN, so q_N is a nonnegative real polynomial of degree at most $2mN$, with $q_N(0) = 1$.

Fix $M > 0$ and let $C_0, \alpha, \varepsilon_0 > 0$ be the constants of Proposition 2.1. If Z_N is empty, the proposition excludes $s_{\min}(T_N(H^{(N)})) \le \varepsilon$ on $\max_j \|H_j^{(N)}\| \le M$ for $0 < \varepsilon \le \varepsilon_0$. Otherwise, on the same event the proposition gives

$$s_{\min}(T_N(H^{(N)})) \le \varepsilon \quad \Longrightarrow \quad \operatorname{dist}(g, Z_N) \le C_0\sqrt{N}\,\varepsilon^{\alpha}.$$

The factor $\sqrt{N}$ converts the Hilbert–Schmidt distance in the original matrix coordinates to distance in the standard Gaussian coordinates. Apply Lemma 3.1 with $D := 2mN$ and $\rho := C_0\sqrt{N}\,\varepsilon^{\alpha}$. Since $\sqrt{n} = \sqrt{d}\,N$, the resulting probability bound is

$$\mathbb{P}\{s_{\min}(T_N(H^{(N)})) \le \varepsilon,\ \max_j \|H_j^{(N)}\| \le M\} \le CN^{5/2}\varepsilon^{\alpha}\exp(CN^{5/2}\varepsilon^{\alpha}) + e^{-cN^2}.$$

Take $M > 2$, so that the norm event has probability tending to one, and choose $A > 0$ with $A\alpha > 5/2$. The right side tends to zero for $\varepsilon = N^{-A}$, proving (1.2). □

4. From the pencil estimate to Brown measure

4.1. **Linearization.** We prove Theorem 1.2. For constant P, both measures in the theorem are the same point mass. Assume henceforth that P is nonconstant, and set $A_N := P(H^{(N)})$ and $a := P(s)$.

Fix $z \ne P(0)$. We represent $P - z$ as a Schur complement of an affine pencil, using one block for each word of length at least two (compare [Han26, Section 2.2]). For a term $c_w w$ with $w = x_{i_1}\cdots x_{i_k}$ and $k \ge 2$, let V_w be the $(k-1)\times(k-1)$ matrix with entries $(V_w)_{r,r+1} := x_{i_{r+1}}$ for $1 \le r \le k-2$ and zeros elsewhere. Set

$$Q_w := I_{k-1} - V_w, \qquad B_w := (c_w x_{i_1}, 0, \dots, 0), \qquad C_w := (0, \dots, 0, -x_{i_k})^{\mathsf{T}}.$$

The polynomial inverse $Q_w^{-1} = I_{k-1} + V_w + \cdots + V_w^{k-2}$ gives $-B_w Q_w^{-1} C_w = c_w w$. Let Q be the block diagonal matrix with blocks Q_w, and assemble the B_w as a block row B and the C_w as a block column C in the same order. With a_z the affine part of $P - z$, the pencil

$$L_z(x) := \begin{pmatrix} a_z & B \\ C & Q \end{pmatrix}$$

has Schur complement $a_z - BQ^{-1}C = P - z$. If P is affine we simply take $L_z := P - z$.

Write q for the size of L_z and let $D_z := L_z(0) = \operatorname{diag}(P(0) - z, 1, \dots, 1)$. Left multiplication by $D_z^{-1} \otimes I_N$ gives a monic pencil

$$\widetilde{T}_N(z) := (D_z^{-1} \otimes I_N)L_z(H^{(N)}) = I_q \otimes I_N + \sum_{j=1}^{d} \widetilde{R}_j(z) \otimes H_j^{(N)}.$$

The upper left $N \times N$ block of its inverse, when it exists, is $(P(0) - z)(A_N - zI_N)^{-1}$. Taking operator norms yields

$$s_{\min}(A_N - zI_N) \ge |P(0) - z|\, s_{\min}(\widetilde{T}_N(z)). \tag{4.1}$$

This also holds in the singular case because $\widetilde{T}_N(z)$ is invertible exactly when $A_N - zI_N$ is invertible. For the fixed shift z, Theorem 1.3 and (4.1) give a $B_z > 0$ such that

$$\mathbb{P}\{s_{\min}(A_N - zI_N) < N^{-B_z}\} \longrightarrow 0. \tag{4.2}$$

For each N, $|\det(A_N - zI_N)|^2$ is a real polynomial in the Gaussian coordinates and is nonzero at the zero input tuple. Its zero set has Lebesgue measure zero, and thus probability zero.

4.2. **The logarithmic tail.** For each fixed $z \neq P(0)$, set $Q_z := (P-z)^*(P-z)$. Let ν_N^z be the empirical law of the squared singular values of $A_N - zI_N$, and let ν^z be the spectral distribution of $Q_z(s)$ with respect to τ. Write

$$G_N(w) := \int \frac{1}{w-x}\, d\nu_N^z(x), \qquad G(w) := \int \frac{1}{w-x}\, d\nu^z(x).$$

The scalar polynomial Q_z is self-adjoint and nonconstant. Indeed, if P_k is the top homogeneous part of P, then the top homogeneous part of Q_z is $P_k^* P_k$. For words u, v of length k, each word u^*v determines both u and v, so a word u with nonzero coefficient in P_k contributes a nonzero coefficient to u^*u.

The resolvent comparison of [CGP22, Corollary 1.3 of arXiv v1], applied to Q_z with scalar coefficients, gives

$$|\mathbb{E}G_N(w) - G(w)| \le C_z N^{-2}((\Im w)^{-5} + (\Im w)^{-2}), \qquad \Im w > 0. \tag{4.3}$$

By [SS15, Theorem 1.1(5)–(6) and Corollary 1.2], G has a convergent expansion in fractional powers at zero and ν^z has no atoms. The bound $|G(\mathrm{i}t)| \le t^{-1}$ excludes powers below -1 along the positive imaginary axis. Also,

$$\mathrm{i}tG(\mathrm{i}t) = \int \frac{\mathrm{i}t}{\mathrm{i}t - x}\, d\nu^z(x) \longrightarrow \nu^z(\{0\}) = 0.$$

Thus the expansion has no term of order -1 either, and for some $0 < \beta_z \le 1$,

$$t|\Im G(\mathrm{i}t)| \le C_z t^{\beta_z}, \qquad 0 < t \le 1.$$

Using $\mathbf{1}_{[0,t]}(x) \le 2t^2/(x^2+t^2)$ for $x \ge 0$ and (4.3), we obtain

$$\mathbb{E}\nu_N^z([0,t]) \le C_z(t^{\beta_z} + N^{-2}t^{-4}), \qquad 0 < t \le 1.$$

Choose $0 < \gamma_z < 2/(4+\beta_z)$ and write $t_N := N^{-\gamma_z}$. The resolvent error can then be absorbed into the first term for $t \ge t_N$, giving

$$\mathbb{E}\nu_N^z([0,t]) \le C_z t^{\beta_z}, \qquad t_N \le t \le 1. \tag{4.4}$$

The same kernel inequality gives $\nu^z([0,t]) \le C_z t^{\beta_z}$ for all $0 < t \le 1$.

On the event $s_{\min}(A_N - zI_N) \ge N^{-B_z}$, the contribution to the negative logarithm from $0 < x \le t_N$ is at most $2B_z(\log N)\nu_N^z([0,t_N])$. The expectation of this bound tends to zero by (4.4). For $t_N < x < \varepsilon < 1$, write $-\log x = |\log \varepsilon| + \int_x^\varepsilon dt/t$. Taking expectations and applying (4.4) gives

$$\mathbb{E}\int_{(t_N,\varepsilon)} |\log x|\, d\nu_N^z(x) \le C_z\varepsilon^{\beta_z}|\log\varepsilon| + C_z\int_{t_N}^{\varepsilon} t^{\beta_z - 1}\, dt \le C_z'\varepsilon^{\beta_z}(1 + |\log\varepsilon|)$$

for all sufficiently large N at each fixed ε. Markov's inequality and (4.2) now give, for every $\delta > 0$,

$$\lim_{\varepsilon\downarrow 0}\limsup_{N\to\infty} \mathbb{P}\left\{\int_{(0,\varepsilon)} |\log x|\, d\nu_N^z(x) > \delta\right\} = 0. \tag{4.5}$$

The bound on $\nu^z([0,t])$ gives a vanishing logarithmic tail for ν^z by the same integration formula.

4.3. **Convergence of potentials.** The GUE moment convergence and norm bounds from [CGP22, pp. 1–2 of arXiv v1] give weak convergence of ν_N^z to ν^z in probability. In fact, normalized trace moments converge, and both measures are supported in a common fixed compact interval with probability tending to one. Polynomial approximation on that interval gives convergence against bounded continuous functions. Consequently we may first truncate the logarithm to a bounded continuous function, with a fixed upper cutoff above that interval. Then (4.5) and the limiting tail bound let us send the lower cutoff to zero. We conclude that, for every fixed $z \neq P(0)$,

$$u_N(z) := \frac{1}{N} \log |\det(A_N - zI_N)| = \frac{1}{2} \int \log x \, d\nu_N^z(x) \xrightarrow{\mathbb{P}} u_a(z) = \tau(\log |a - z1_{\mathcal{M}}|). \tag{4.6}$$

To take the distributional Laplacian, we show local L^1 convergence in probability. Choose $M > \|a\|$ so that $\mathbb{P}\{\|A_N\| \leq M\} \to 1$, and let $K \subset \mathbb{C}$ be compact. On this norm event, the eigenvalues of A_N belong to the closed disk of radius M. The functions $z \mapsto \log|z-w|$ have uniformly bounded $L^2(K)$ norm for $|w| \leq M$, since the translated sets $K - w$ lie in one bounded disk and $\log|\cdot|$ is locally square integrable. Using the determinant formula $u_N(z) = N^{-1} \sum_{k=1}^N \log|z - \lambda_k(A_N)|$ and Jensen's inequality gives $\|u_N\|_{L^2(K)} \leq C_{M,K}$ on the norm event. The potential representation of the Brown measure gives $\|u_a\|_{L^2(K)} \leq C_{M,K}$ as well.

For $\delta > 0$, let $E_{N,\delta} := \{z \in K : |u_N(z) - u_a(z)| > \delta\}$, taking the potentials up to changes on planar null sets. Fubini's theorem, (4.6) and dominated convergence give $\mathbb{E}|E_{N,\delta}| \to 0$, where $|\cdot|$ denotes planar Lebesgue measure. On the norm event, Cauchy–Schwarz yields

$$\|u_N - u_a\|_{L^1(K)} \leq \delta|K| + 2C_{M,K}|E_{N,\delta}|^{1/2}.$$

Since the norm event has probability tending to one and δ is arbitrary, this proves $u_N \to u_a$ locally in L^1 in probability.

For $\varphi \in C_c^\infty(\mathbb{C})$, the distributional identity $\Delta \log|z| = 2\pi\delta_0$ and Definition 1.1 give

$$\int \varphi \, d\mu_{A_N} = \frac{1}{2\pi} \int_{\mathbb{C}} u_N \Delta\varphi \, d^2z \xrightarrow{\mathbb{P}} \frac{1}{2\pi} \int_{\mathbb{C}} u_a \Delta\varphi \, d^2z = \int \varphi \, d\mu_a^{\mathrm{Br}}.$$

On the norm event both probability measures are supported in the disk of radius M. Smooth compactly supported functions approximate any bounded continuous test function uniformly on this disk. This gives the convergence in Theorem 1.2.

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